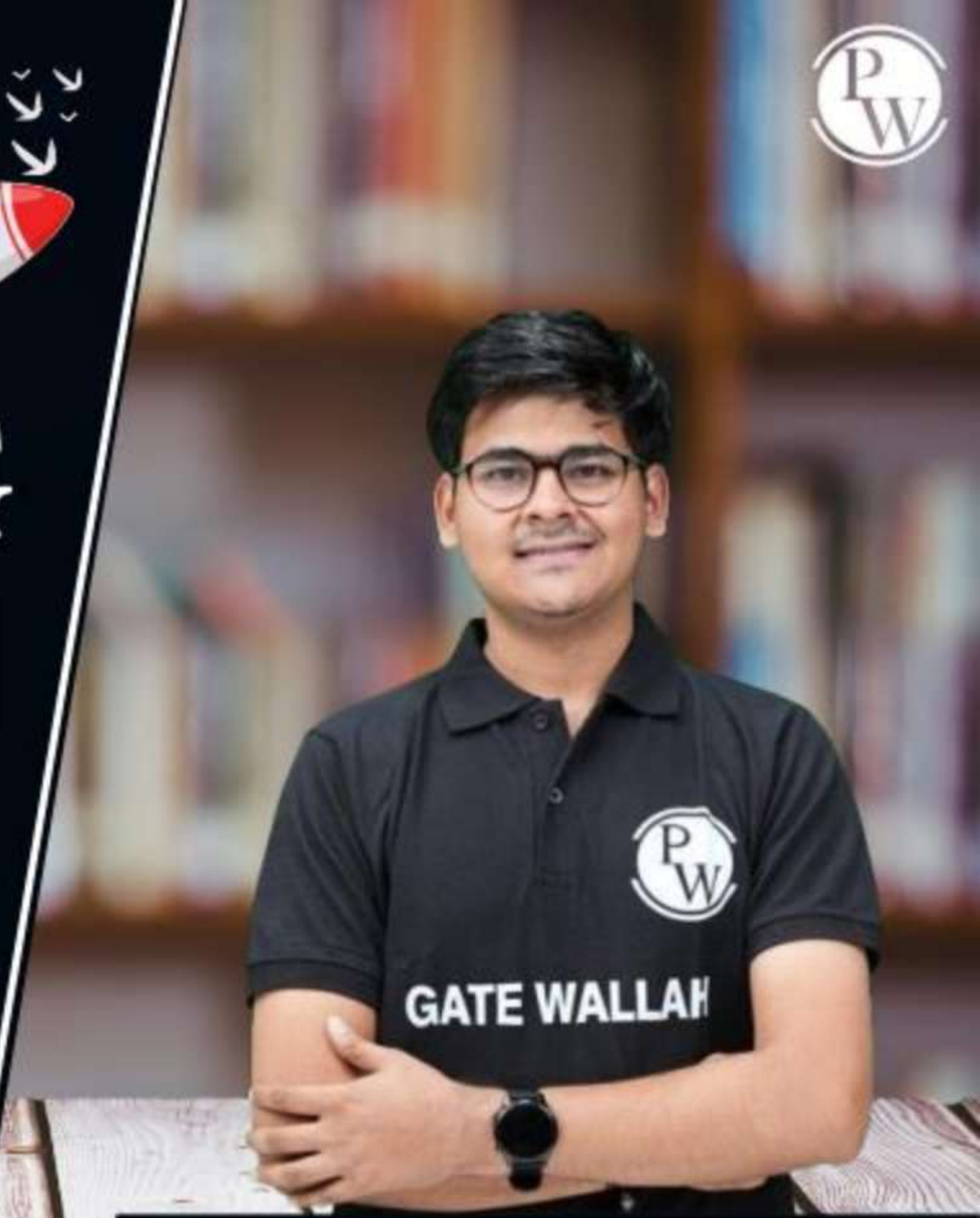
CS & IT ENGINEERING

Digital Logic

Optimisation

Lecture No. 02



By- Aditya sir

Recap of Previous Lecture

Topic

Universal Gates

NAND, NOR

Topics to be Covered

Topic

Continue with Universal Gates

Practice Questions

Case 1 :- $y = A \cdot B \cdot \overline{C} \cdot D$

NAND :- $(2n-2)+k$

NOR :- $(3n-3)-k$

n = total no. of i/p variables

k = no. of compl var.

Case 2 :- $y = A + \overline{B} + C + D + E$

NAND : $(3n-3) - k$

NOR : $(2n-2) + k$

(g) $y = \overline{P} + \overline{Q} + \overline{R}$

min. NAND $\longrightarrow a = 3$

$a + b = ?$

min. NOR $\longrightarrow b$

$\longrightarrow 7$

$7 + 3 = 10$

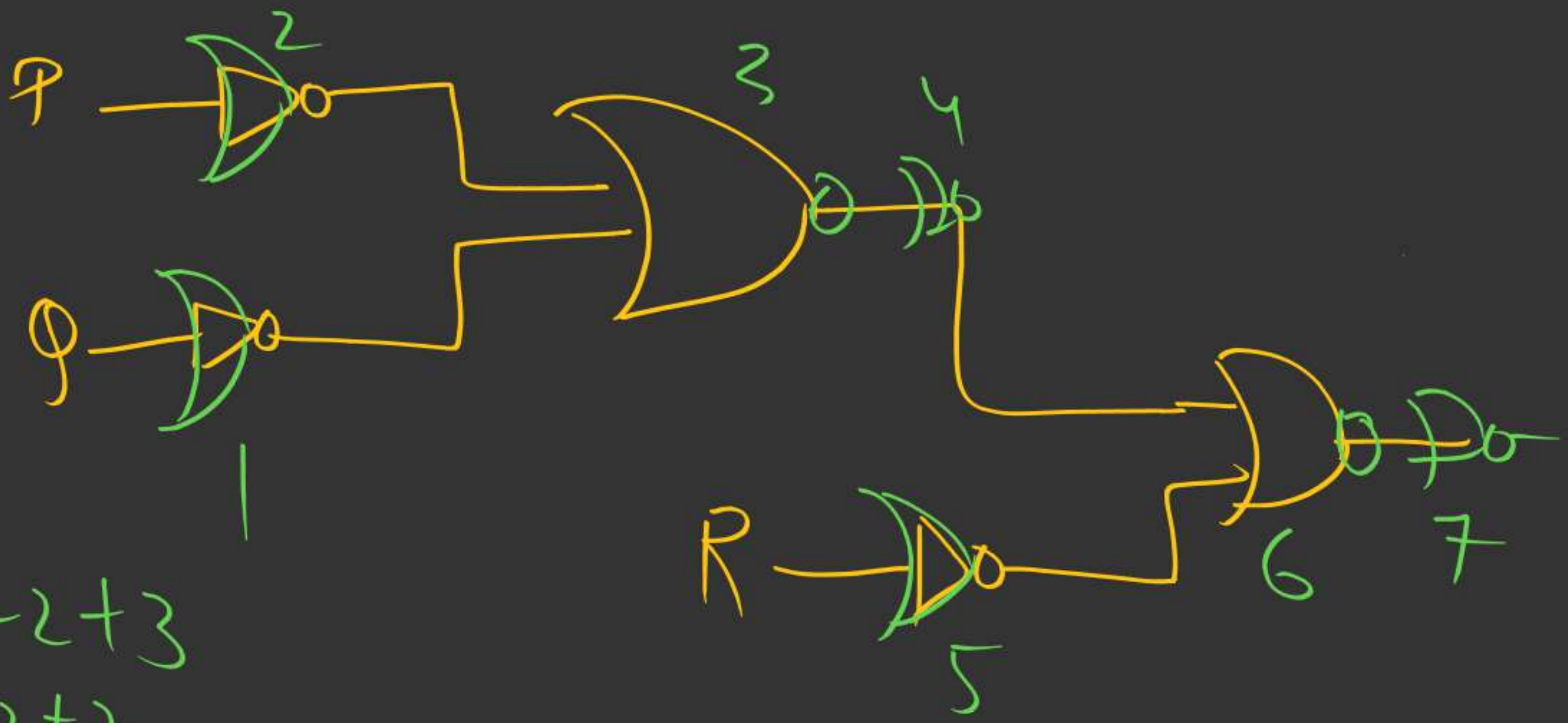
$y = \overline{P} + \overline{Q} + \overline{R}$

$b = 7$

NOR

$\downarrow$

$(2n-2)+K$



$n=3$

$K=3$

$3\times2-2+3$

$6-2+3$

$=7$

$\underline{\underline{a=3}}$

$n=3, k=3$

$3n-3-k$

$3\times3-3-3$

$=3$

$$y = \overline{P} \cdot \overline{Q}\, R \cdot \overline{S}$$

min. NAND $\rightarrow$ a $\rightarrow$ 9

min. NOR $\rightarrow$ b $\rightarrow$ 6

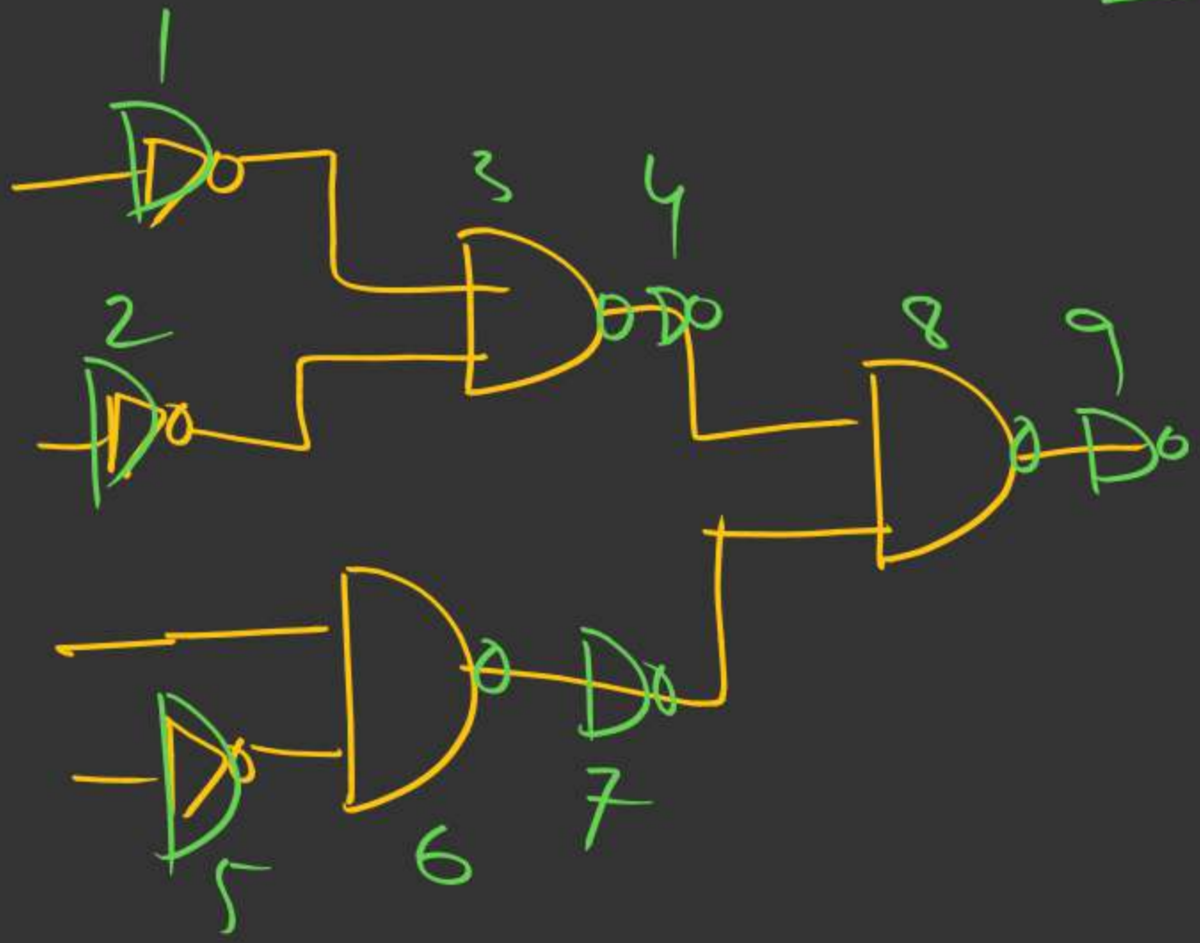
$a * b = ?$

$9 \times 6 = \underline{\underline{54}}$

~~52%~~

$y = \overline{P}\,\overline{Q}\,R\,\overline{S}$

NAND



$a = 9$

Cal :- $(2n-2)+k$

$n=4$
$k=3$

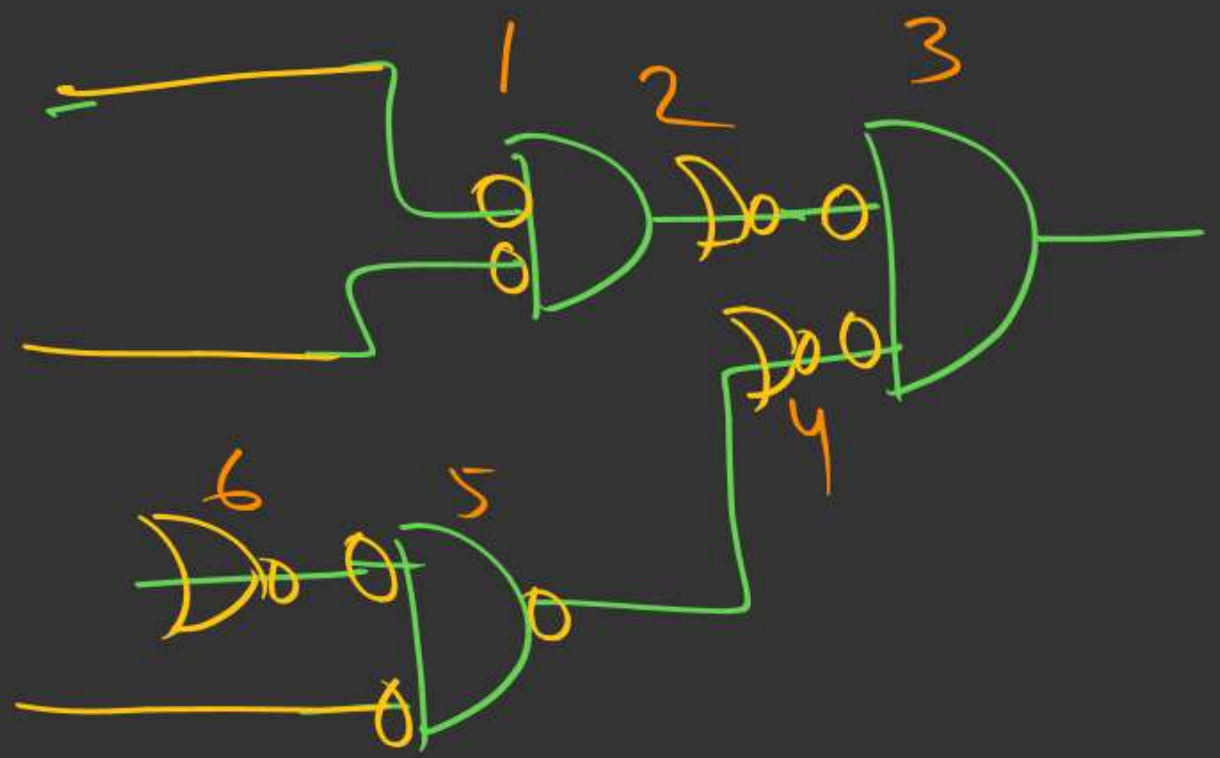
$2\times4-2+3$

$= 8+1 = 9$

NOR

$y = \bar{P}\bar{Q}R\bar{S}$

$b = 6$



$n = 4, k = 3$

$3n - 3 - k$

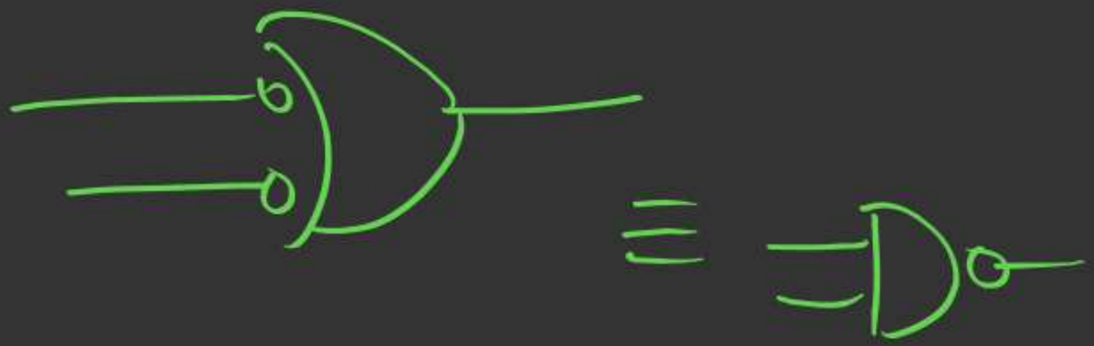
$3 \times 4 - 3 - 3$

$= 12 - 6 = 6$

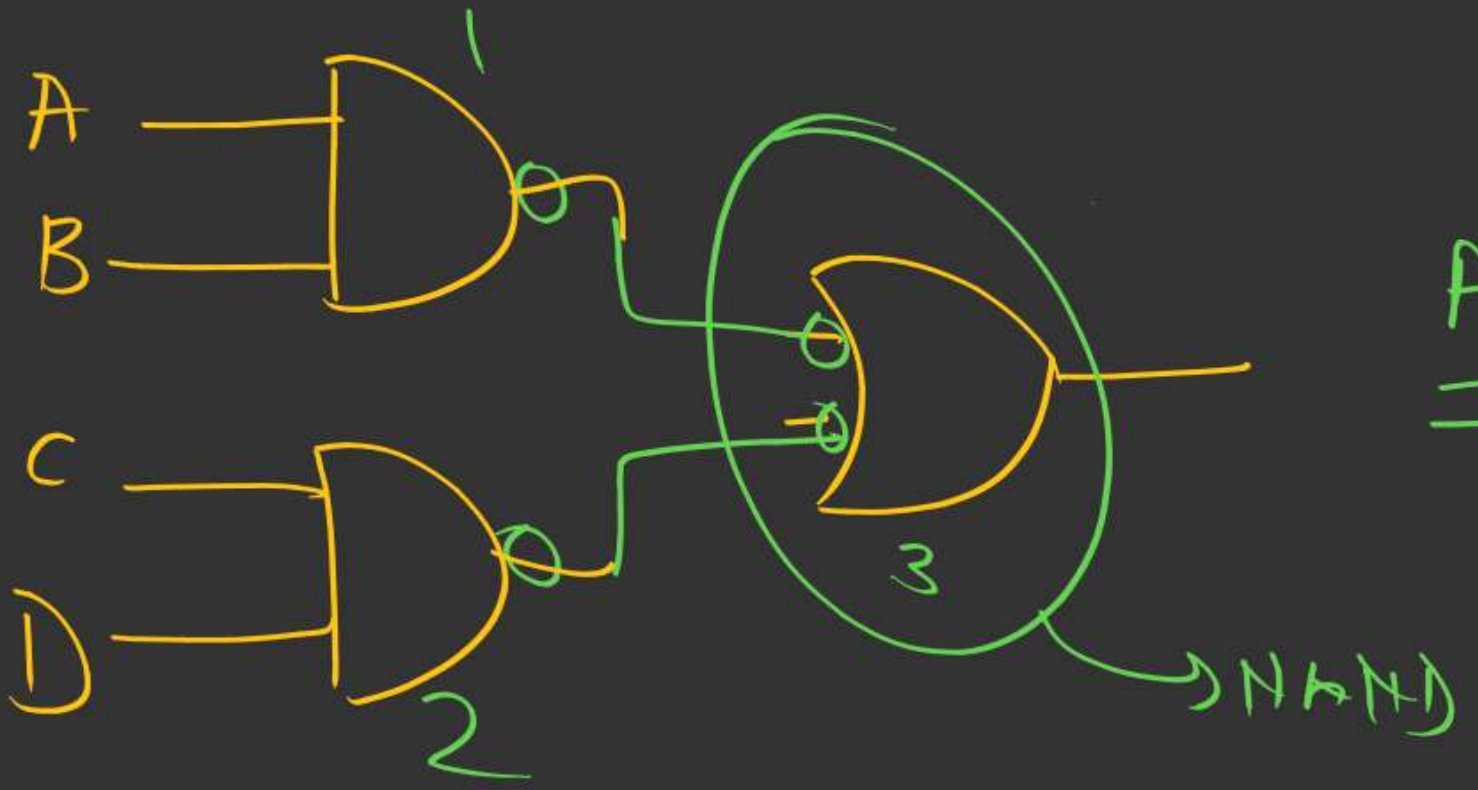
Case 3:- SOP format

NAND

$y = AB + CD$



AOI



Ans: 3

SOP $\rightarrow$ AB+CD

AND AND

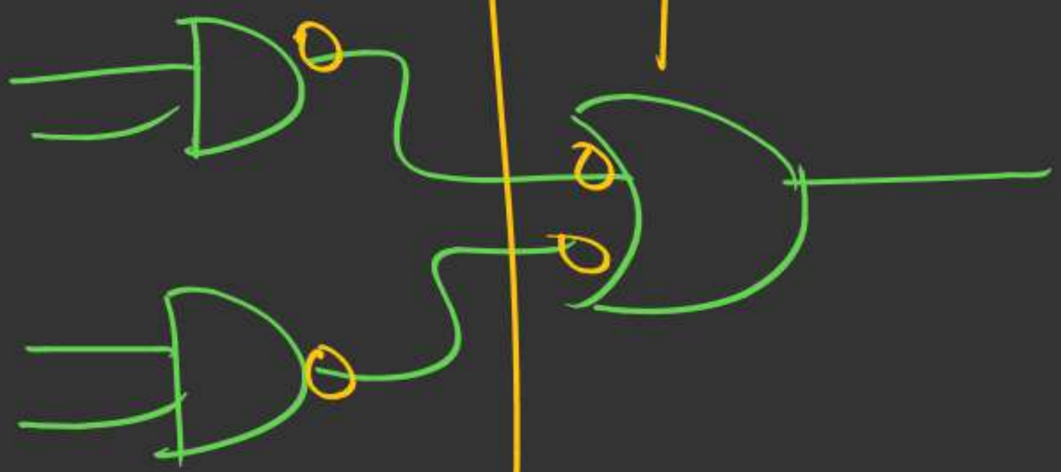
OR

AOI = NAND

$\downarrow$

AB+CD

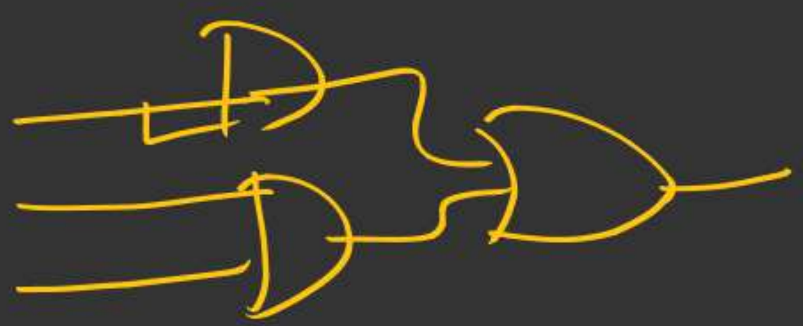
AND $\leftrightarrow$ OR



AND $\rightarrow$ OR

NAND $\rightarrow$ NAND

Imp:- → Min. no. of NAND



Steps:-

1) Write → SOP format
2) Use AOI ⟹ $\boxed{[\text{AND} \rightarrow \text{OR}]}$
3) Ans is exactly equal to AOI

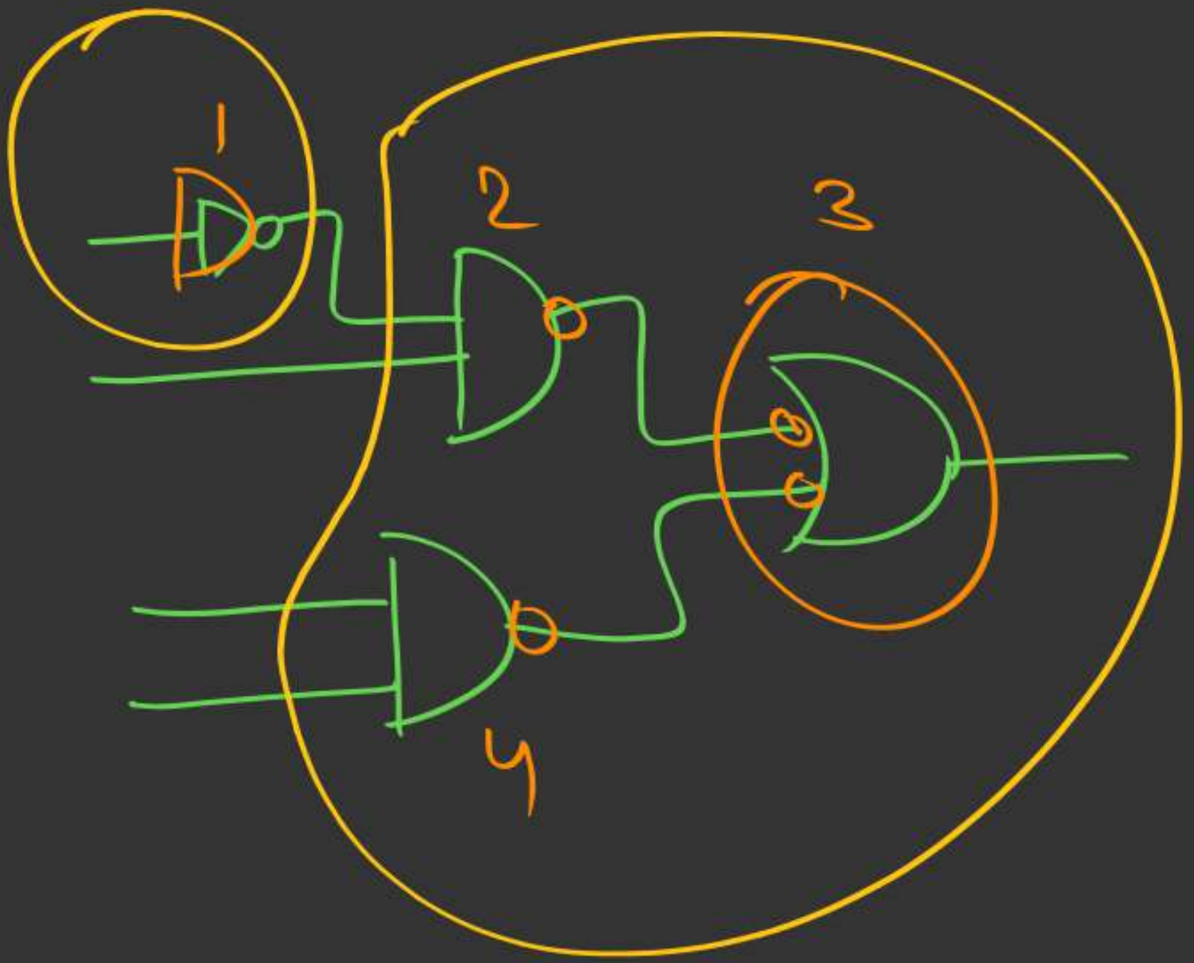
bcoz AND → OR = (NAND → NAND)

Check diagram

$\underline{Q92)}$:- $y = \bar{P}Q + RS \longrightarrow \underline{SOP}$

62%

NAND



AJ Shortcut

$y = \bar{P}Q + RS$

Let $\bar{P} = A$ → 1 NAND

$y = AQ + RS$ → 3 NAND

$3+1 = 4$

71%

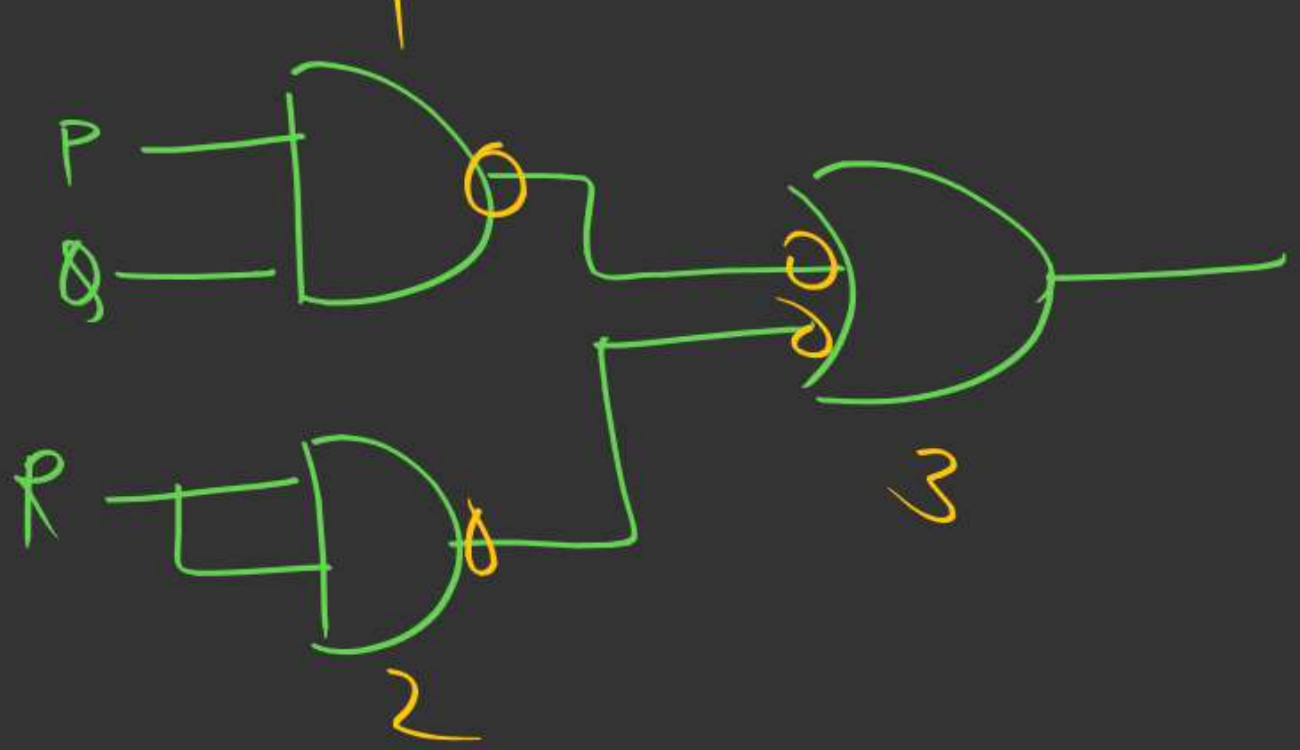
Q3)

$y = PQ + R$

$PQ + RS$

NAND?

$y = PQ + R \cdot R$



Ans: 3

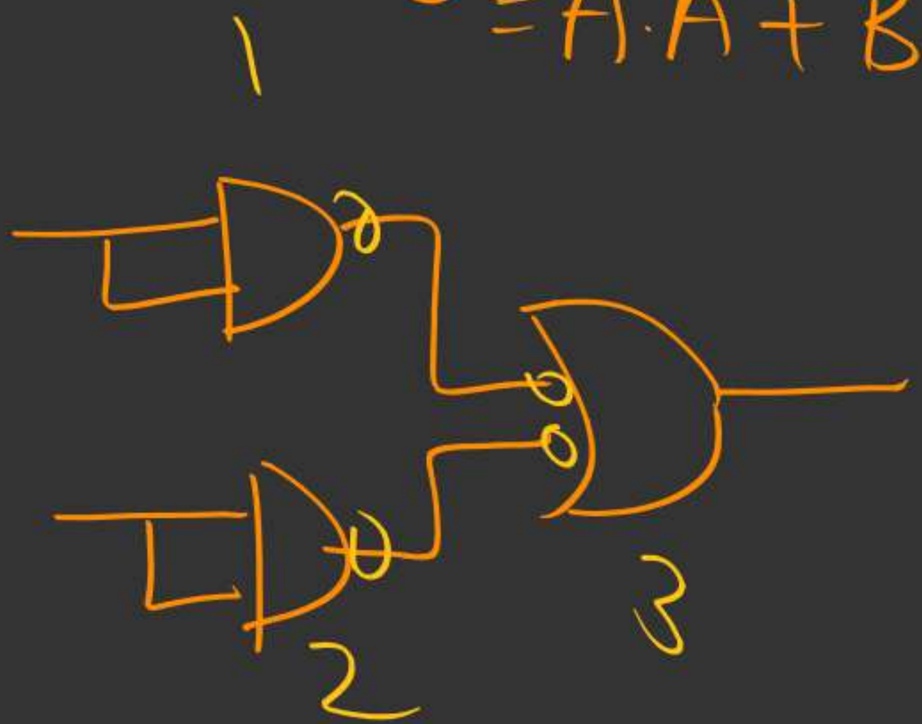
PQ+RS

eg4)

$y = A + B$

$y = A + B$
$= A \cdot A + B \cdot B$

Solve using Case 3



3 NAND

Q5) $y = PQ + RS + AB$

min NAND ?

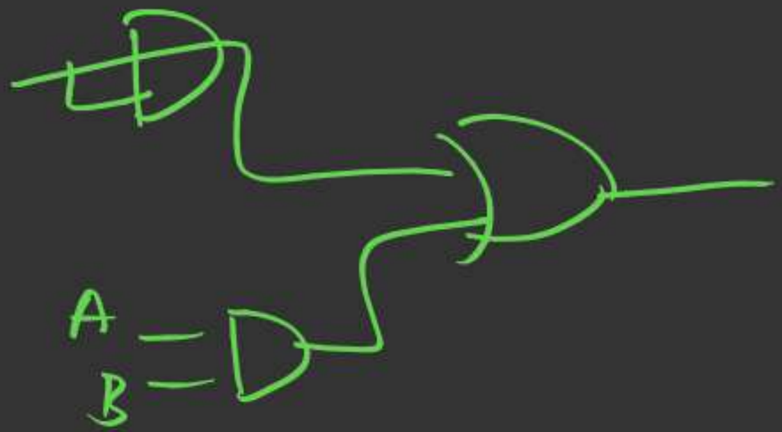
3 NAND

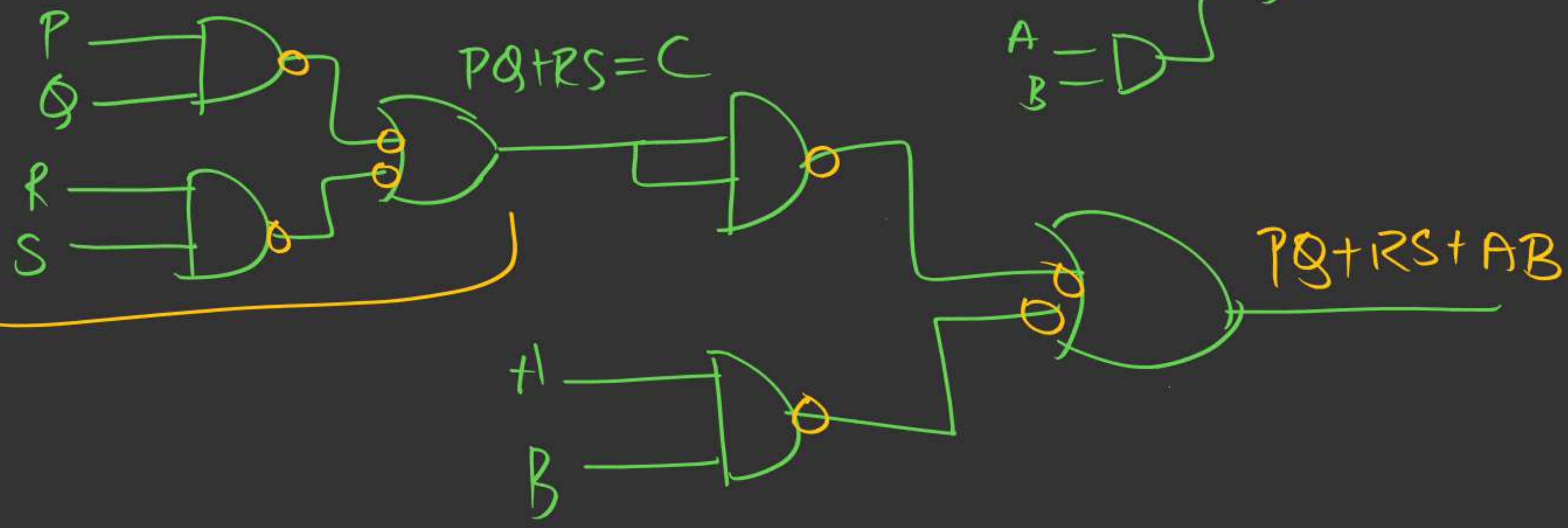
AND → OR

70%

45%

2 i/p





Explanation

$$y = \underbrace{PQ + RS}_{} + AB$$

$PQ + RS$

Let $C = PQ + RS$ → 3 NAND

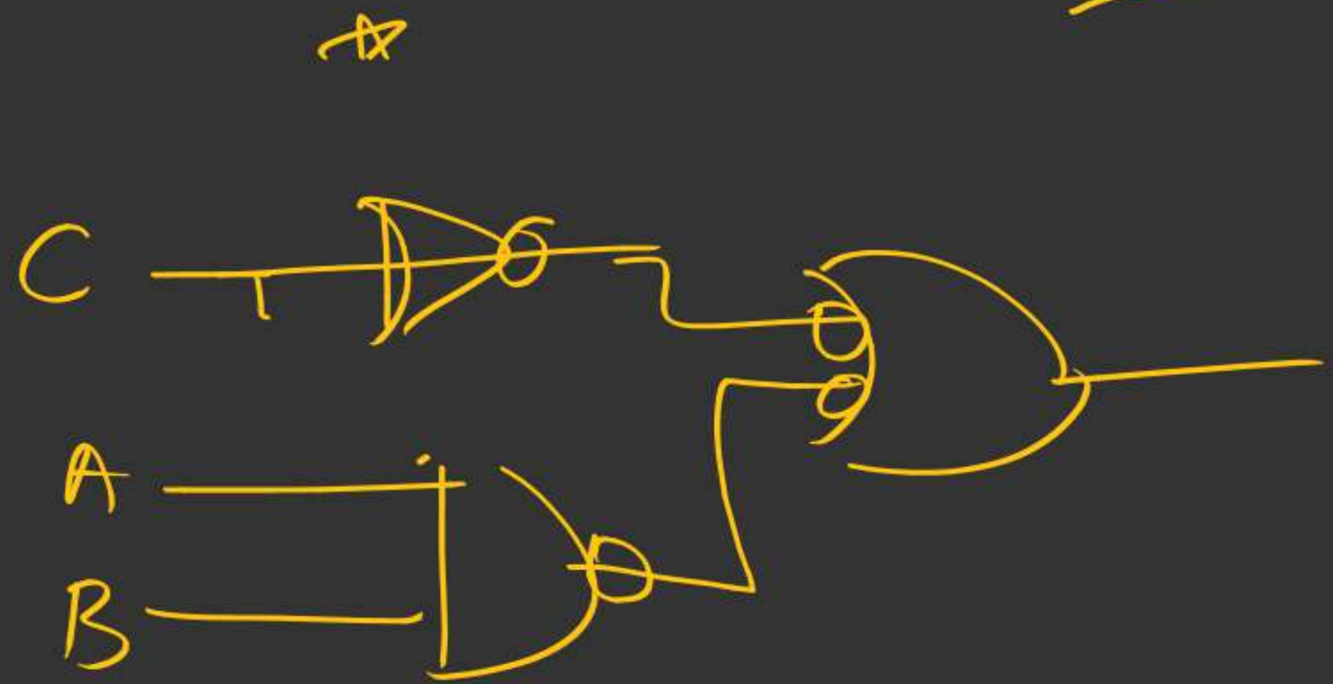
$$y = C + AB$$

$$y = C \cdot C + AB$$

3 NAND

$3 + 3 = 6$ NAND

min NAND



76%

$y = PQ + R + AB$

$= \underbrace{PQ + AB}_{C} + R$

min NAND = ?

$C \rightarrow$ 3 NAND

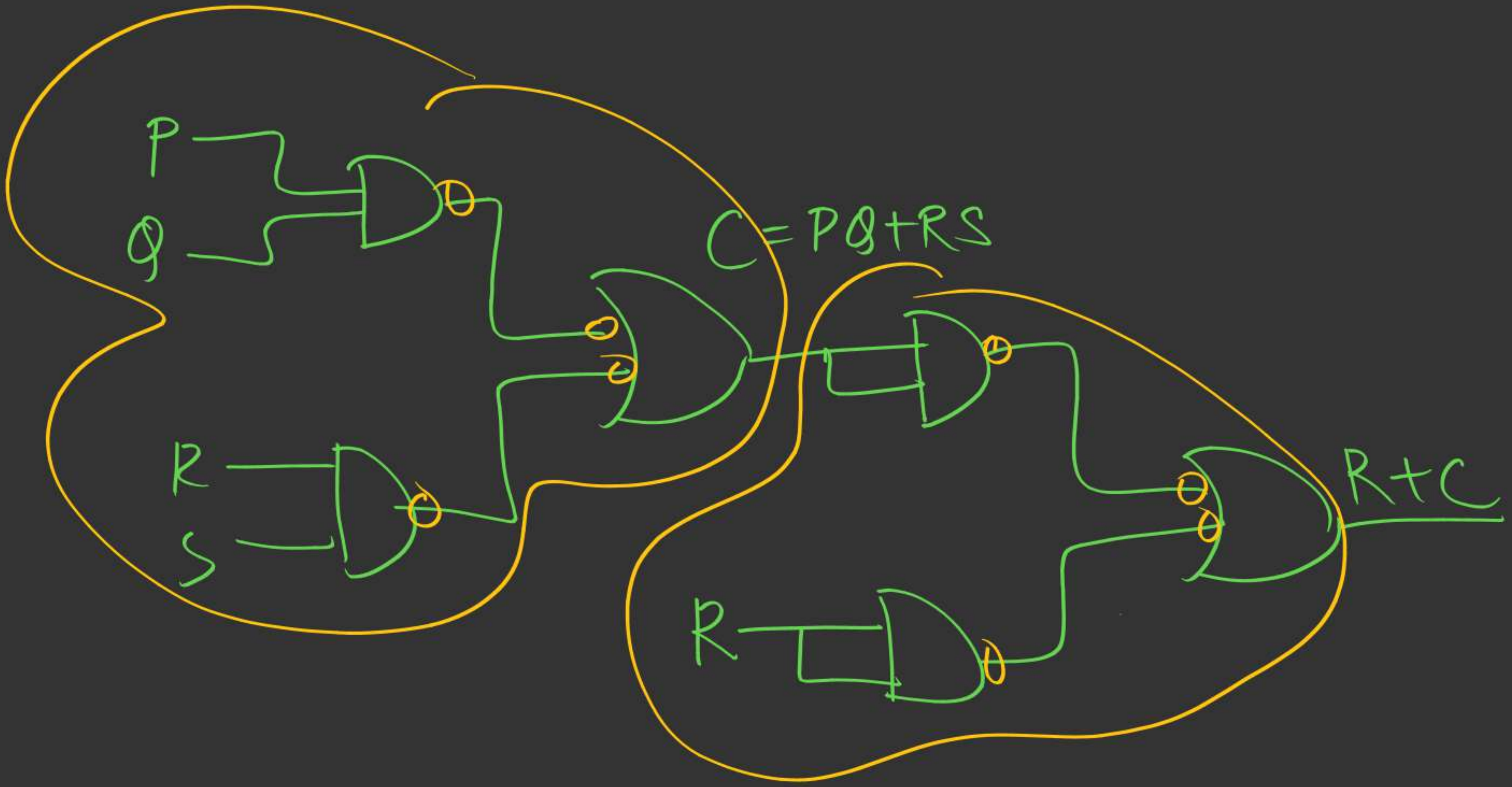
$C = PQ + RS \rightarrow$ 3 NAND

$y = C + R$

$C \cdot C + R \cdot R$ } 3 NAND

6 NAND

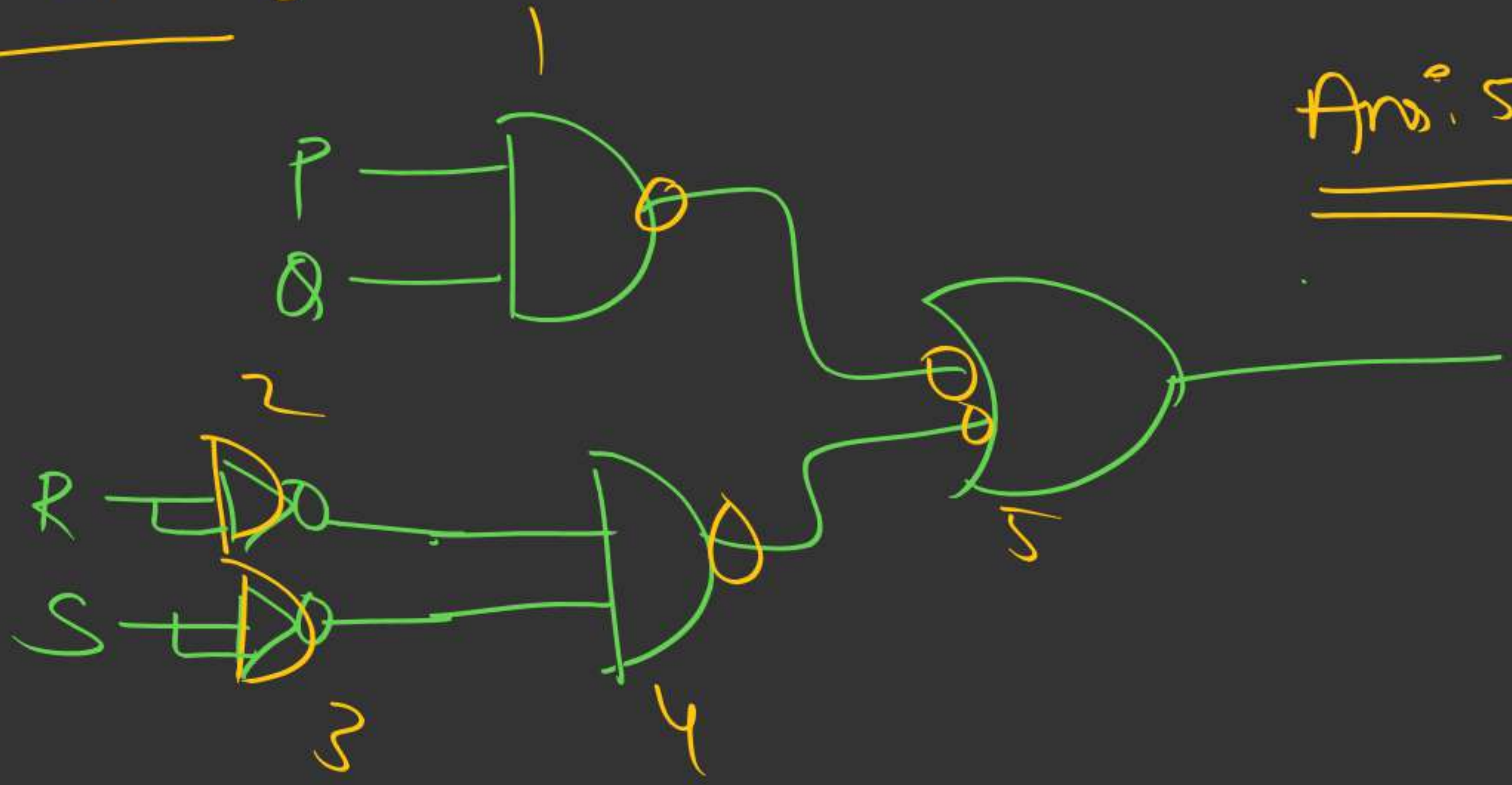
6 NAND



70%

$$y = PQ + \bar{R}\bar{S}$$

min NAND = ?



Ans: 5

$y = PQ + \bar{R}\,\bar{S}$

Let $A = \bar{R} \longrightarrow$ 1 NAND

$B = \bar{S} \longrightarrow$ 1 NAND

$y = PQ + AB \longrightarrow$ 3 NAND

$\longrightarrow 1+1+3 = 5$

10%

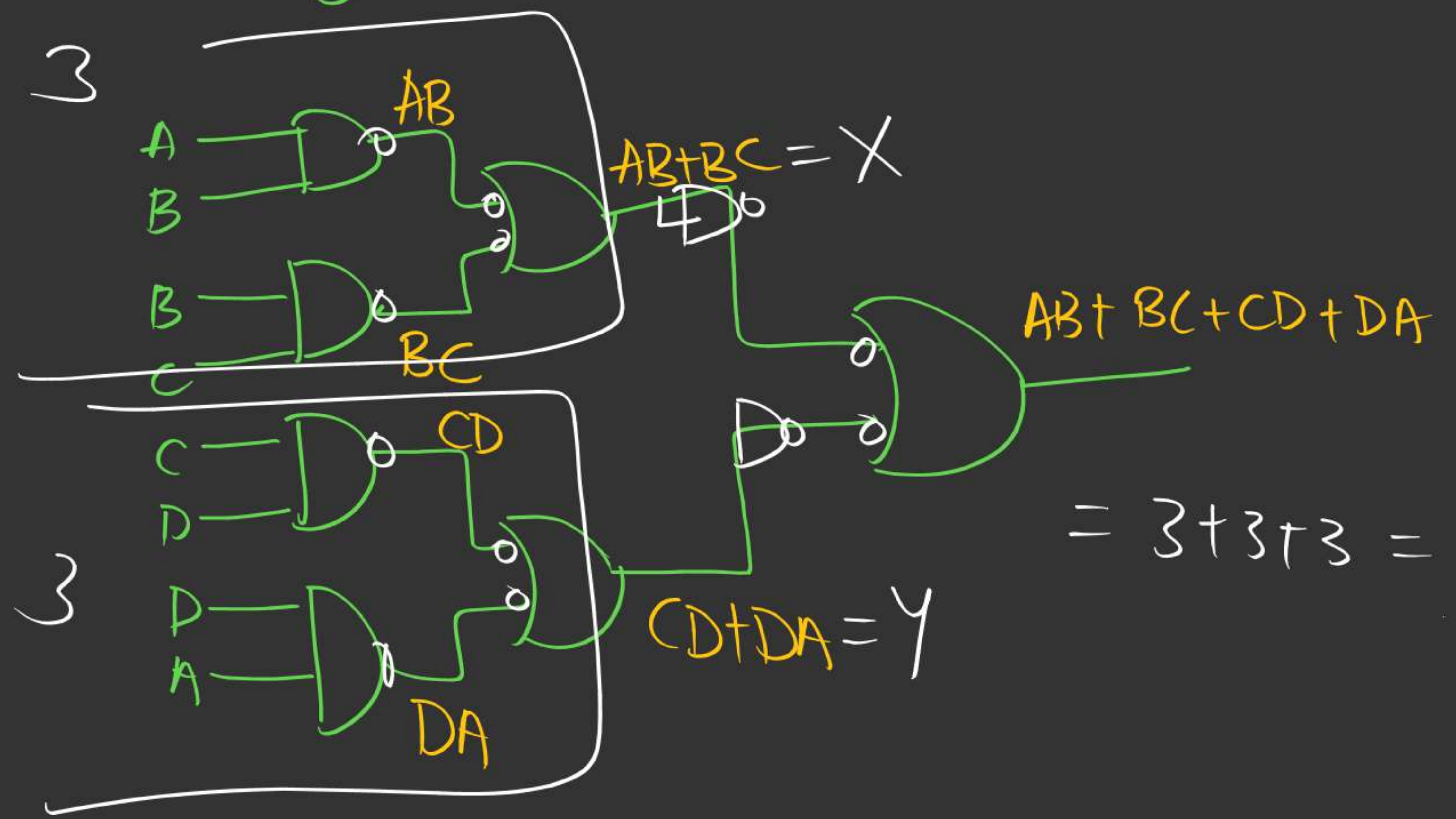
$$y = AB + BC + CD + DA$$

min NAND = ?

9

8

$$y = AB + BC + CD + DA$$



$$= 3+3+3 = 9$$

$$y = \underbrace{AB + BC}_{P} + \underbrace{CD + DA}_{Q}$$

$$P = AB + BC \longrightarrow 3$$

$$Q = CD + DA \longrightarrow 3$$

$$3 + 3 + 3 = 9$$

$$y = P + Q$$

$$= P \cdot P + Q \cdot Q \longrightarrow 3$$

Appr 2 :-

$$y = AB + BC + CD + DA$$

$$= B(A+C) + D(C+A)$$

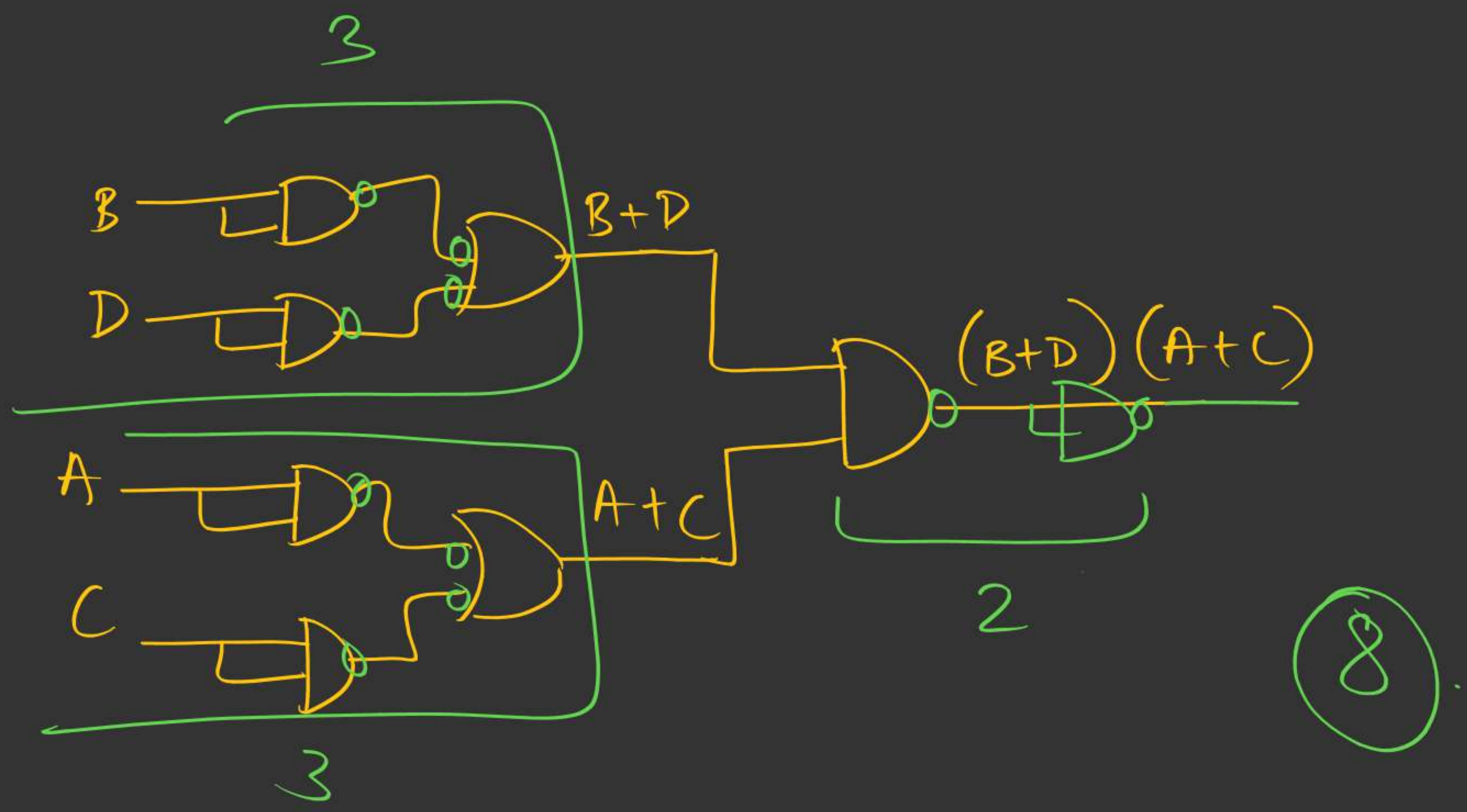
$$= (B+D)(A+C)$$

Let $P = B + D \longrightarrow 3$

$Q = A + C \longrightarrow 3$

$y = P \cdot Q \longrightarrow 2$

$3 + 3 + 2 = 8$


3
B
D
B+D
(B+D)(A+C)
A
C
A+C
2
3
8

$$y = AB + BC + CD + DA$$

$$y = B(A+C) + D(A+C)$$

☆ Let $\boxed{P = (A+C)} \rightarrow$ 3 NAND

$$y = B \cdot P + D \cdot P$$

$\rightarrow$ 3 NAND

$3 + 3 = \underline{\underline{6}}$

H.W 1 $\longrightarrow$ Draw diagram

2 mins Summary

NAND , NOR

THANK - YOU